

Name: _____ Partner: _____ Period: _____

The Engineering Design Challenge

Engineers use function models to make decisions. In each scenario below, you will **construct a model** and **apply it** to answer design questions. Work through your assigned tier. If you finish early, try the next tier.

Tier R — Remediate

Scenario: A rectangular solar panel array has one side against a wall. The available border is 60 m. The area function has already been set up for you:

$$A(w) = -2w^2 + 60w, \quad 0 < w < 30$$

where w is the width in meters and A is the area in square meters.

1. Evaluate the model at each width. Show your substitution.

w (m)	Work	$A(w)$ (m ²)
5		_____
10		_____
15		_____
25		_____

2. Compute the average rate of change of A on the interval $[5, 15]$.

$$\text{ARC} = \frac{A(15) - A(5)}{15 - 5} = \frac{\text{_____}}{\text{_____}} = \text{_____}$$

Units: _____

3. Based on your table in part (1), between which two widths does the maximum area appear to occur?

Between $w = \text{_____}$ m and $w = \text{_____}$ m.

Tier A — Apply

Scenario: A storage company charges based on distance from a central depot. Testing shows that the shipping cost C (in dollars) is inversely proportional to the square of the distance d (in km) from the depot, with proportionality constant $k = 2000$.

1. Write the rational function model $C(d)$.

$$C(d) = \underline{\hspace{10cm}}$$

2. State the domain restriction and explain it in context.

Domain: _____ because _____

3. Complete the table and describe the trend.

d (km)	2	4	10	20
$C(d)$ (\$)	_____	_____	_____	_____

Trend: As distance increases, cost _____.

4. Compute the average rate of change of C on $[2, 4]$. Include units.

$$\text{ARC} = \frac{C(4) - C(2)}{4 - 2} = \frac{\quad}{\quad} = \underline{\hspace{2cm}}$$

Units: _____

Tier E — Extend

Scenario: An engineer collects three data points for a project's profit P (in thousands of dollars) versus number of units produced x (in hundreds):

x	1	2	3
$P(x)$	-1	5	5

- Assume $P(x) = ax^2 + bx + c$. Substitute each data point to write a system of three equations in a , b , c .

$(1, -1)$: _____

$(2, 5)$: _____

$(3, 5)$: _____

- Solve the system. Show your elimination or substitution steps.

$a =$ _____ $b =$ _____ $c =$ _____

- Write the complete quadratic model.

$P(x) =$ _____

- State one assumption this model makes that may not hold in reality.

- Predict the profit when $x = 4$ (400 units). Is this prediction inside or outside the range of the data?

$P(4) =$ _____ thousand dollars. This is _____ the data range.